Introduction to Operations Research, Deterministic Models

Lecture 07: Integer Programming Models

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Integer programming (IP)

Applications of IP involves:

- the extension of LP where the divisibility assumption is dropped:
  - the units of products must be integer
  - the number of labor involvement must be integer
- a number of interrelated "yes-or-no" decisions:
  - Should we undertake a particular fixed project?
  - Should we make a particular fixed investment?
  - Should we locate a facility in a particular site?

$$x_j = \begin{cases} 1 & \text{if decision } j \text{ is yes} \\ 0 & \text{if decision } j \text{ is no} \end{cases}$$

Such variables are called binary variables or 0-1 variables.

Binary integer programming (BIP): IP problems that contain only binary variables. They are also called as **0–1 integer programming** problems.

Pure integer programming: all variables are integers

Mixed integer programming: only some variables need to be integers

A prototype example

The CALIFORNIA MANUFACTURING COMPANY is considering expansion by building a new factory in either Los Angeles or San Francisco, or perhaps even in both cities. It also is considering building at most one new warehouse, but the choice of location is restricted to a city where a new factory is being built.

Decision variables:

$x_1 \in \{0, 1\}$ represents whether to build a factory in Los Angeles,
$x_2 \in \{0, 1\}$ represents whether to build a factory in San Francisco,
$x_3 \in \{0, 1\}$ represents whether to build a warehouse in Los Angeles,
$x_4 \in \{0, 1\}$ represents whether to build a warehouse in San Francisco.

The net present value and capital required are shown below. The objective is to find the feasible combination of alternatives that maximizes the total net present value.

Yes or No question	decision variable	net present value	capital required
whether to build a factory in LA	x_1	9 million	6 million
whether to build a factory in SF	x_2	5 million	3 million
whether to build a warehouse in LA	x_3	6 million	5 million
whether to build a warehouse in SF	x_4	4 million	2 million

A prototype example

In the prototype example:

- Building at most one new warehouse: $x_3 + x_4 \leq 1$
- The choice of warehouse location is restricted to a city where a new factory is being built: $x_3 \leq x_1, x_4 \leq x_2$.

Common relations among "yes-or-no" decisions:

- mutually exclusive alternatives such that only one decision in the group can be "yes".
- one decision is contingent on another decision:
  - a decision is allowed to be "yes" only if the other decision is "yes"
  - a decision must be "yes" if the other decision is "yes".
  - a decision must be "no" if the other decision is "yes".

Capital Budgeting Problems

Capital budgeting problem

There are four investments and each investment must be either fully invested or not invested.

- Investment 1: Invest: 5,000; Return: 16,000
- Investment 2: Invest: 7,000; Return: 22,000
- Investment 3: Invest: 4,000; Return: 12,000
- Investment 4: Invest: 3,000; Return: 8,000

With the total budget of 14,000, the goal is to maximize the total return from all investments.

Decision variables:

$$x_i = \begin{cases} 1 & \text{if investment } i \text{ is invested} \\ 0 & \text{else} \end{cases} \quad \text{for } i = 1, 2, 3, 4.$$

Capital budgeting problem

BIP problem:

$$\begin{aligned} \text{maximize} \quad & 16x_1 + 22x_2 + 12x_3 + 8x_4 \\ \text{subject to} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14, \\ & x_i \in \{0, 1\}, \forall 1 \leq i \leq 4. \end{aligned}$$

Extensions:

- Can only invest in at most two investments:
- If investment 2 is invested, investment 1 must also be invested:
- If investment 1 is invested, investment 2 cannot be invested:

Investments in general: Management often must face decisions about whether to make fixed investments (those where the amount of capital required has been fixed in advance).

- Should we purchase a certain source of raw materials?
- Should we add a new production line to produce a certain input item ourselves?

General form of Capital budgeting

In a typical capital-budgeting problem, decisions involve the selection of potential investments.

The investment decisions might be to choose among possible plant locations, or to select a configuration of capital equipment. Often it makes no sense to consider partial investments in these activities, and so the problem becomes an integer program, where the investment decision variables are taken to be $x_j = 0$ or 1.

Assuming that c_j is the contribution resulting from the jth investment and that $a_{i,j}$ is the amount of resource i, such as cash or manpower, used on the jth investment, we can state the problem formally as:

$$\begin{aligned} \text{maximize} \quad & \sum_{j=1}^{n} c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i, \forall 1 \leq i \leq m \\ & x_j \in \{0,1\}, \forall 1 \leq j \leq n. \end{aligned}$$

The capital-budgeting model can be made much richer by including logical considerations.

A special capital budgeting problem: Knapsack problem

When there is only one resource constraint, the capital budgeting problem reduces to

$$\begin{aligned} \text{maximize} \quad & \sum_{j=1}^{n} c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^{n} a_j x_j \leq b \\ & x_j \in \{0,1\}, \forall 1 \leq j \leq n. \end{aligned}$$

This problem is called the **0–1 knapsack problem**, since it is analogous to a situation in which a hiker must decide which goods to include in his backpack on this trip.

Here c_j is the "value" or utility of including good j, which weighs $a_j > 0$ pounds; the objective is to maximize the "pleasure of the trip" subject to the weight limitation that the hiker can carry no more than b pounds.

The model is altered somewhat by allowing more than one unit of any good to be taken, by writing $x_j \geq 0$ and x_j integer in place of the 0–1 restrictions on the variables. (e.g., the subproblem in the cutting stock example)

Production and Distribution Network design

Production and Distribution Network design

- **Site selection:** Before selecting a site for a new plant, many potential sites may need to be analyzed and compared. Each of the potential sites involves a "yes-or-no" decision to decide whether a certain site should be selected.

- **Fixed cost:** such a charge occurs when a production run is undertaken and the required production facilities must be set up to initiate the run.

- **Shipment dispatch:** daily operating decisions need to be made about how to send the shipments. For instance, it involves the "yes or no" selection of a certain route, with a certain truck size, that departs at a certain time period.

Warehouse location problem

There are two supply locations (S_1, S_2) with given supply capacity of 50 and 75 respectively and two stores (D_1, D_2) with given annual demand requirements of 75 and 50 respectively. The plan is to construct a warehouse at one of three sites (A, B, C). products can be shipped from supply locations to warehouses, warehouses to demand points, supply points to demand points.

- Potential warehouses have annual capacity and fixed cost.
- The firm has developed a transport cost matrix.

warehouse	capacity	fix cost
A	∞	500
B	60	720
C	70	680

Distance	S_1	S_2	A	B	C
A	1	6			
B	2	3			
C	8	1			
D_1	4	7	4	3	5
D_2	8	6	6	4	3

- Formulate a mixed integer LP to minimize the total cost

Warehouse location problem

Let x_1, x_2 and x_3 denote the binary decision variable representing whether the site A, B, or C is selected to construct a warehouse.

Let $z_{1,k}$ and $z_{2,k}$ represent the unit number that shipped from supply locations S_1 and S_2 to the three potential warehouse locations for $k = 1, 2, 3$ respectively

Let $y_{k,1}$ and $y_{k,2}$ represent the unit number that shipped from the three potential warehouse locations to the two stores for $k = 1, 2, 3$ respectively

- only one warehouse is to be constructed: $x_1 + x_2 + x_3 \leq 1$
- shipment is allowed to site A/B/C only if a warehouse is constructed at that site: $Mx_k \geq z_{1,k} + z_{2,k}$, for $k = 1, 2, 3$.
- the total cost is $500x_1 + 720x_2 + 680x_3 +$ shipment costs

More generally, the total cost of the activity is the sum of a variable cost that is proportional to the level of the activity and the setup cost required to initiate the activity.

Fixed cost in MIP

- With x_j denotes the level of activity j ($x_j \geq 0$), $k_j(\geq 0)$ denotes the setup cost, and c_j denotes the cost for each incremental unit, the total cost of the activity (say, activity j) can be represented by a function of the form $f_j(x_j) = \begin{cases} k_j + c_j x_j & \text{if } x_j > 0 \\ 0 & \text{else} \end{cases}$.

- Adding an auxiliary binary variable y_j which represents whether activity j should be undertaken, then $Z = \sum_j f_j(x_j) = \sum_j (k_j y_j + c_j x_j)$ with $y_j = \begin{cases} 1 & \text{if } x_j > 0 \\ 0 & \text{else} \end{cases}$

- Let M be an extremely large positive number that exceeds the maximum feasible value of any x_j, the minimization of Z is then equivalent to

$$\begin{array}{ll} \text{minimize} & \sum_j (k_j y_j + c_j x_j) \\ \text{subject to} & M y_j \geq x_j,\ x_j \geq 0,\ \forall j \\ & y_j \in \{0, 1\}, \forall j \end{array}$$

Facility location problem

Suppose that a manager must decide which of n warehouses to use for meeting the demands of m customers for a product.

- The demand d_j of each customer j must be filled from the warehouses.
- Goods can be shipped from a warehouse only if it is opened. Let k_i be the fixed operating cost of warehouse i.
- Let the transportation cost from i to j be $c_{i,j}$.

Traveling salesman problem

Consider a salesman who must visit six cities.

- Must start from city 1 and return to city 1.
- Must visit each city exactly once
- Every two cities i and j are connected with the distance d_{ij} for $i, j \in \{1, \ldots, 6\}$
- Formulate the problem as a mixed-integer linear programming problem to to minimize the total distance traveled.

Decision variables: binary variables x_{ij} denote whether to visit city j after visiting city i.

- $x_{ii} = 0$, $\sum_{j=1}^{6} x_{ij} = 1$, $\sum_{i=1}^{6} x_{ij} = 1$ for $i \in [6], j \in [6]$
- Subtour elimination: for all $S \subseteq \{2, 3, 4, 5, 6\}$, $\sum_{i \in S, j \in S} x_{i,j} \leq |S| - 1$.

Airline applications

The airline industry is an especially heavy user of OR throughout its operations.

Fleet assignment problem: Given several different types of airplanes available, the problem is to assign a specific type to each flight leg that leaves from a crew base and returns to the same base in the schedule so as to maximize the total profit from meeting the schedule.

Binary decision variables represent whether one should assign a certain type of airplane to a certain flight leg.

Crew assignment problem: for each feasible sequence of flight legs, the problem is to assign sequences of flight legs to crews of pilots and flight attendants. Moreover, their crew schedules occasionally need to be revised quickly when flight delays or cancellations occur because of inclement weather, aircraft mechanical problems, or crew unavailability.

Binary decision variables represent whether one should assign a certain sequence of flight legs to a certain crew.

Crew assignment problem

SOUTHWESTERN AIR needs to assign its crews based in SF to cover all its upcoming flights given below. The other 12 columns show the 12 feasible sequences of flights for a crew. The cost of assigning a crew to a particular sequence of flights is given in the bottom row.

	feasible sequences of flights											
Flight	1	2	3	4	5	6	7	8	9	10	11	12
SF to LA	1			1			1			1		
SF to Denver		1			1			1			1	
SF to Seattle			1			1			1			1
LA to Chicago				2			2		3	2		3
LA to SF	2					3				5	5	
Chicago to Denver				3	3				4			
Chicago to Seattle							3	3		3	3	4
Denver to SF		2		4	4				5			
Denver to Chicago					2			2			2	
Seattle to SF			2				4	4				5
Seattle to LA						2			2	4	4	2
Costs ($ 1000)	2	3	4	6	7	5	7	8	9	9	8	9

Exactly three of the sequences need to be chosen (one per crew) in such a way that every flight is covered. (It is permissible to have more than one crew on a flight, where the extra crews would fly as passengers, but union contracts require that the extra crews would still need to be paid for their time as if they were working.) The objective is to minimize the total cost of the three crew assignments that cover all the flights.

• **Decision variables:** a binary variable x_j represents whether the sequence j should be assigned a crew, for $j \in [12]$.

• Every flight is covered: $x_1 + x_4 + x_7 + x_{10} \geq 1$ for the Flight SF to LA, and similarly for all other flights.

• Exactly three sequences are assigned the crew: $\sum_{i=1}^{12} x_i = 3$

• **The objective:** $2x_1 + 3x_2 + 4x_3 + 6x_4 + 7x_5 + 5x_6 + 7x_7 + 8x_8 + 9x_9 + 9x_{10} + 8x_{11} + 9x_{12}$

Set covering problem

This crew assignment problem illustrates a broader class of problems called **set covering problems**. Any set covering problem can be described in general terms as involving a number of potential activities (such as flight sequences) and characteristics (such as flights).

Each activity possesses some but not all of the characteristics. The objective is to determine the least costly combination of activities that collectively possess (cover) each characteristic at least once.

Thus, let S_i be the set of all activities that possess characteristic i. At least one member of the set S_i must be included among the chosen activities, so a constraint is $\sum_{j \in S_i} x_j \geq 1$.

A set covering problem is of the form

$$\begin{array}{ll} \text{minimize} & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \sum_{j \in S_i} x_j \geq 1, \forall i = 1, \ldots, p \\ & x_j \in \{0, 1\}, \forall i = j = 1, \ldots, k \end{array}$$

Integers for Constraint Modeling

Disjunctive constraints

- To ensure either $f(x) \leq 0$ or $g(x) \leq 0$ is satisfied,

$$\begin{cases} f(x) \leq My \\ g(x) \leq M(1-y) \\ y \in \{0,1\} \end{cases}$$

with $M \geq \max\left\{\sup_x f(x),\ \sup_x g(x)\right\}$

- If we want to ensure that at least one of the m constraints $\{f_j(x) \leq 0\}_{j\in[m]}$ are satisfied,

$$\begin{cases} f_j(x) \leq M(1-y_j) \\ \sum_{j\in[m]} y_j = 1,\ y_j \in \{0,1\} \end{cases}$$

with $M \geq \max_j \left\{\sup_x f_j(x)\right\}$

If-then constraints

- If $f(x) > 0$ is satisfied, then $g(x) \leq 0$ must be satisfied.
- The above "if-then" constraint could be modeled by $g(x) \cdot \mathbb{I}_{(0,\infty)}(f(x)) \leq 0$
- The above "if-then" constraint is equivalent to either $f(x) \leq 0$ or $g(x) \leq 0$.
- "if-then" constraint is modeled by

$$\begin{cases} f(x) \leq My \\ g(x) \leq M(1-y) \\ y \in \{0,1\} \end{cases}$$

Solving Integer Programming Problems

Solving integer programming problems

- The number of solutions grows in exponential order with the number of integer variables
- Solving general integer programs can be much more difficult than solving linear programs. There is no known polynomial-time algorithm for solving general MILPs.
- For any given IP problem, the corresponding linear programming problem without integer requirements commonly is referred to as its LP relaxation.
- Solving the associated linear programming relaxation results in **an upper bound** on the optimal value to the maximization of MILP.
  - it is optimal to MILP if it satisfies the integer requirements
  - rounding to a feasible integer solution may be difficult
  - the optimal solution to the LP relaxation may be arbitrarily far away from optimality
  - rounding may result in a solution that is not feasible, or far from optimality

Branch and bound methods

Branch and bound is the most commonly-used algorithm for solving MILPs.

General form of MILP

$$\begin{array}{ll} \text{maximize} & c_1^\top x_1 + c_2^\top x_2 \\ \text{subject to} & A_1 x_1 + A_2 x_2 = b \\ & x_1 \geq 0, x_2 \geq 0 \\ & x_1 \text{ integer} \end{array}$$

Let S represent the feasible region of the above MILP problem. With a partition of S into subsets $S_1, \ldots, S_k$ such that $S = S_1 \cup \ldots S_k$. Then

$$\max_{x \in S} \quad c^\top x = \max_{i \in [k]} \left\{ \max_{x \in S_i} c^\top x \right\}$$

In other words, we can optimize over each subset separately.

• Idea: If we can't solve the original problem directly, we might be able to solve the smaller subproblems recursively.

Reading materials

Section 12.1-12.3, *Introduction to Operations Research* by Hillier and Lieberman